

600-12 Volt DC-DC Converter

Team: sdmay26-45

Client: Formula SAE Team

Advisor: Professor Cheng Huang

Members: Caleb Rezmer, Hunter Kanne, Andrew Chandler, Drew Bixler, Diego Rivera, Wesley Mao

Team



Drew
Bixler



Andrew
Chandler



Hunter
Kanne



Diego Rivera



Caleb Rezmer



Wesley Mao

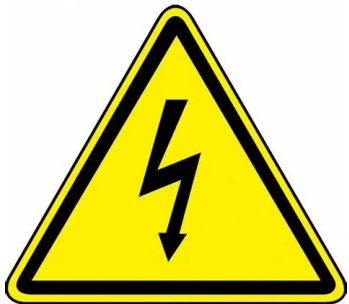
Problem Statement



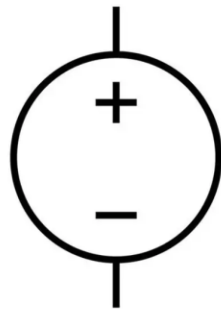
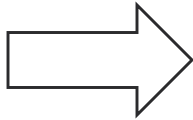
Problem: The Iowa State Formula SAE team is currently designing an electric race car; however, they needed a way to supply substantial power to their low-voltage systems.



Solution: Design an efficient DC-DC converter to power the low-voltage systems from the high-capacity tractive system battery.



600 Volts



12 Volts



Requirements

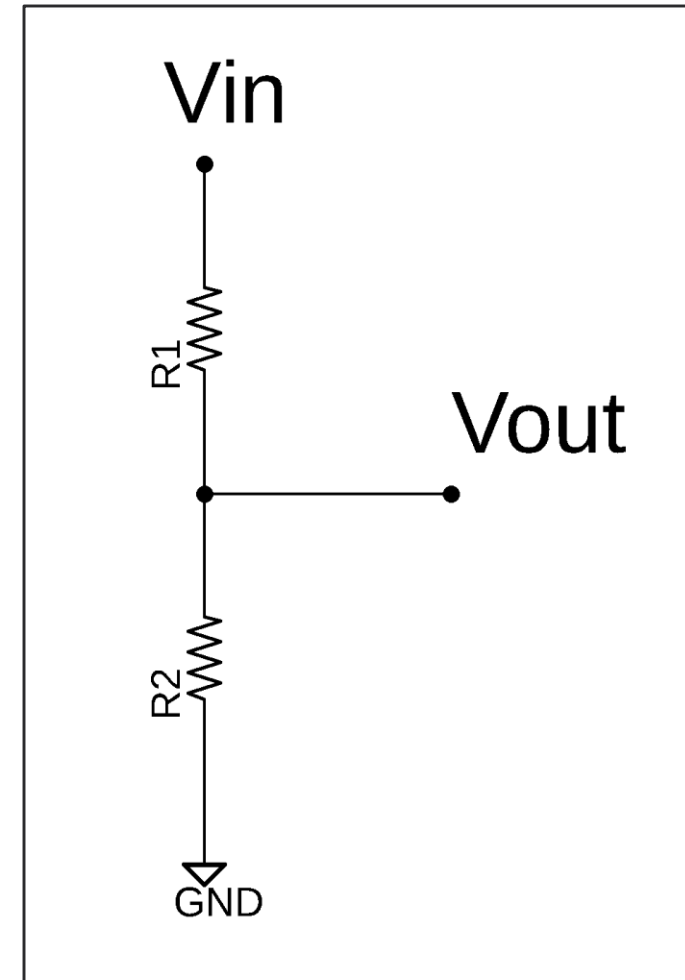


- Input voltage of 400-600 volts
- Stable 12-volt output
- Can supply at least 30 amps of current
- Efficiency > 90%
- Galvanic Isolation
- Complies with FSAE design requirements
 - Voltage separation spacing (4mm)
 - Discharge rates

Approach

After Researching....

- Decided on Half-bridge LLC Converter Topology due to the following factors:
 - Efficiency
 - Galvanic Isolation
 - Wide Load and Input Voltage range
 - Small footprint
 - Interest and complexity (state-of-the-art)
- Found an open-source GitHub repository from a German Formula team that has been developing an LLC converter for several years:
 - Converts 400-600 Vin to 24 volts
 - Maximum output current of 20 amp
- Based on our team's level of experience and confidence in power electronics design, we decided to reference the design and create a 600:12 volt, 30 amp revision



Timeline

Timeline:

9/20

10/21

10/31

11/20

11/30

Research

- Researched DC-DC Converters
- Decide on converter topology

Design

- Discussed design with client and previous team
- Specified new components

Simulation

- Main Simulation
- Precharge Simulation
- Attempted SR simulation

Layout

- Implement new footprints
- Modified existing layout

Ordering

- Created project budget
- Make tradeoffs
- Negotiated cost split
- Acquired sponsor

Implementation

- Tested converter in car
- Added parallel functionality with low voltage battery

Regression

- Adjustments made (Supercap removed)
- Tuned switching frequency

System Testing

- High Voltage Testing
- Load Testing
- Input Range Testing

Unit Testing

- Tested Board With Low Voltage
- Isolation Tested Transformer

Assembly

- Constructed and Tuned Transformer To Specs
- Assembled PCB

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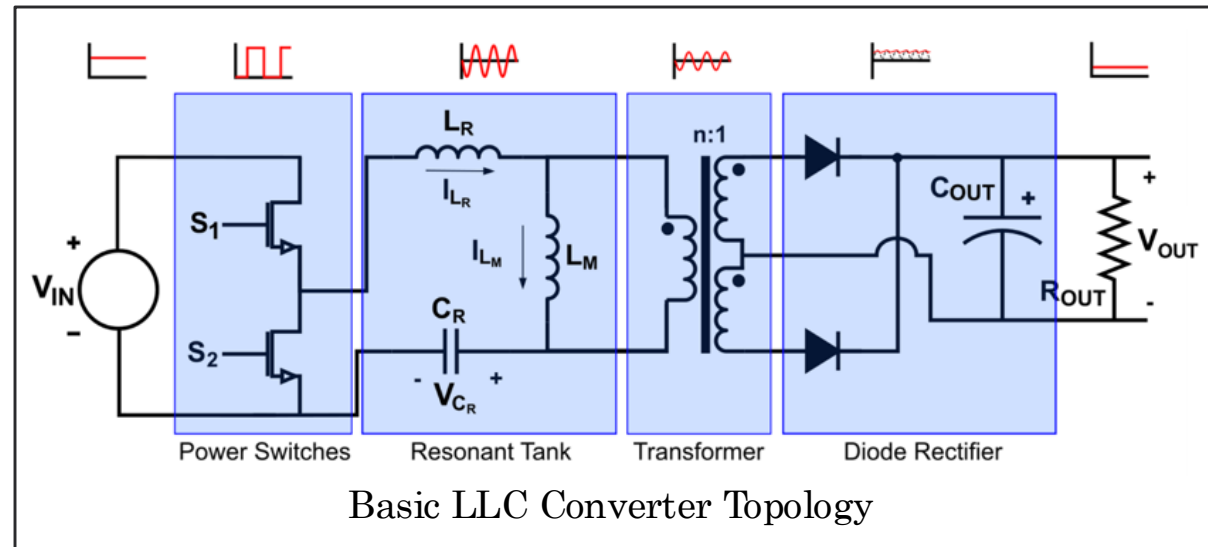
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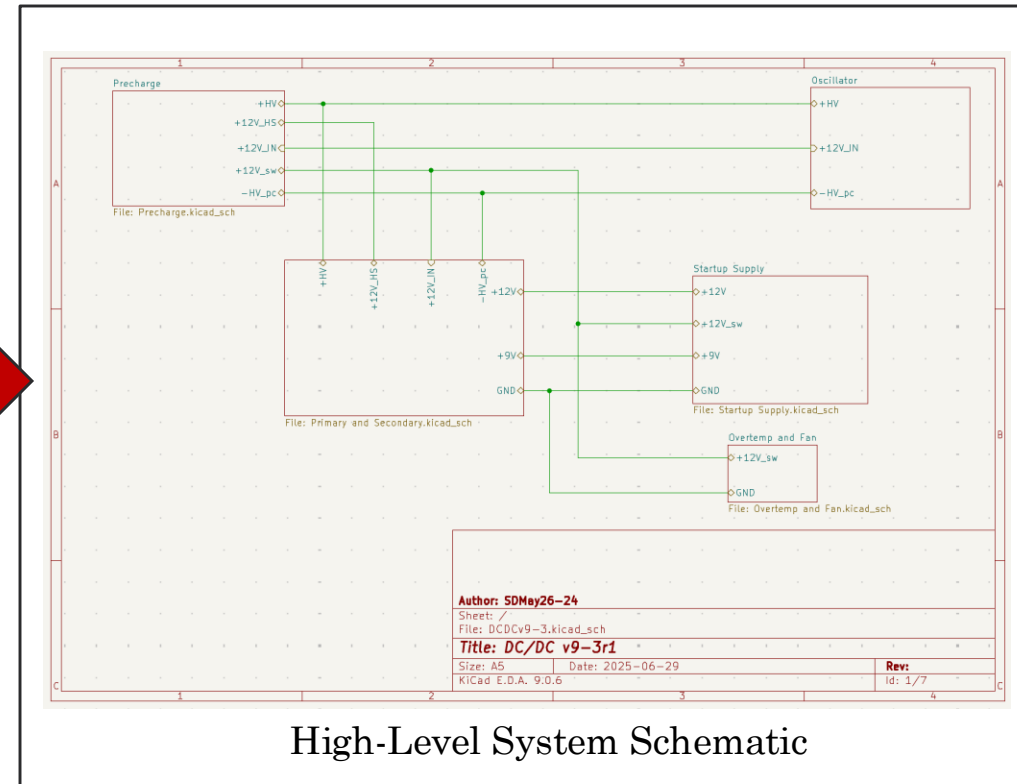
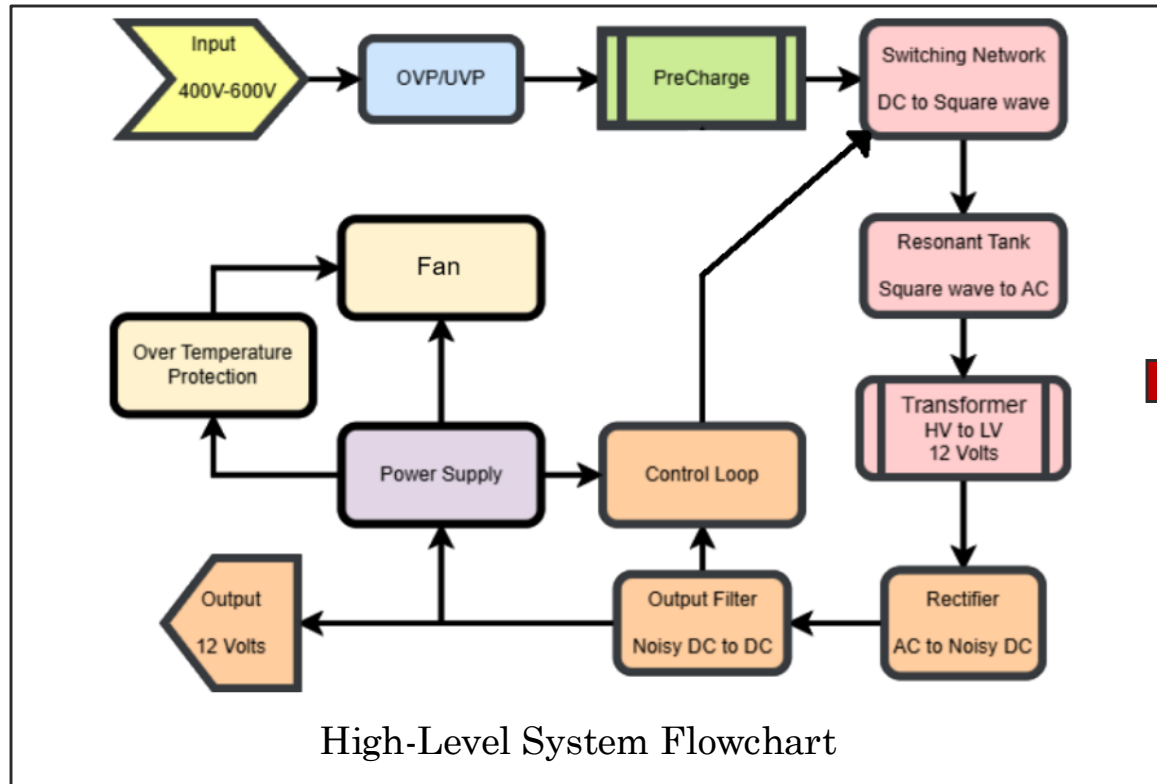
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High-Level Design

- The design of the LLC converter is split up into several subsystems:
 - Pre-Charge
 - Switching Network
 - Resonant tank
 - Transformer
 - Rectifier
 - Output Filter
 - Control Loop
 - Power Supply

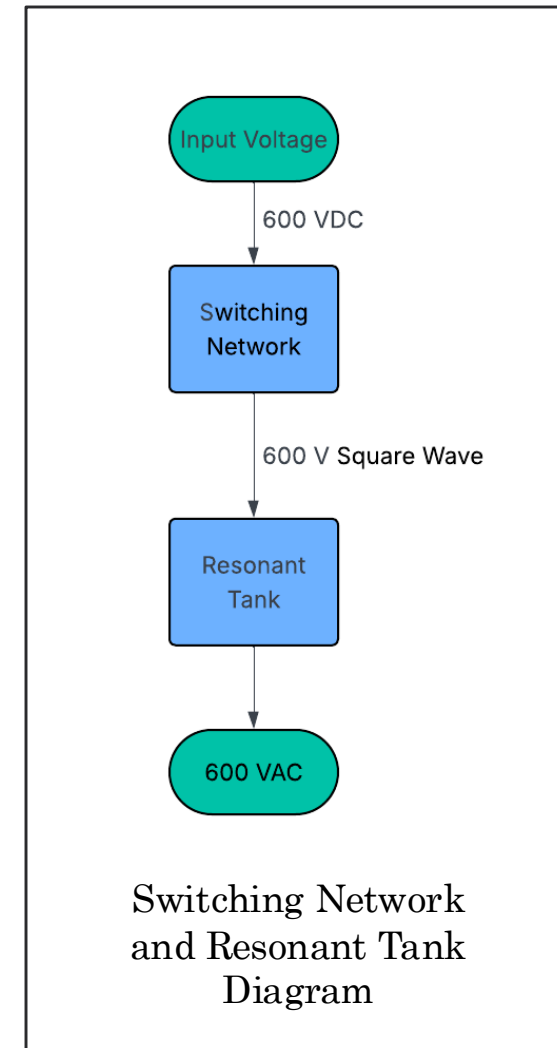


High-Level Design



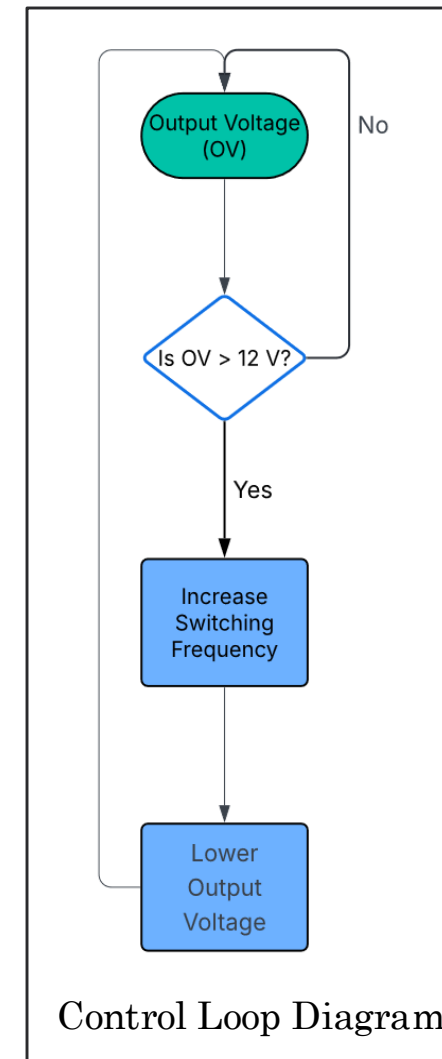
Switching Network and Resonant Tank

- Switching network converts DC input to a square wave
- Resonant tank converts square wave into AC sinusoidal wave
 - This allows transformer to efficiently step 600 V ➡ 12 V
- The minimum switching frequency is set to maximize converter efficiency and effective converter input range



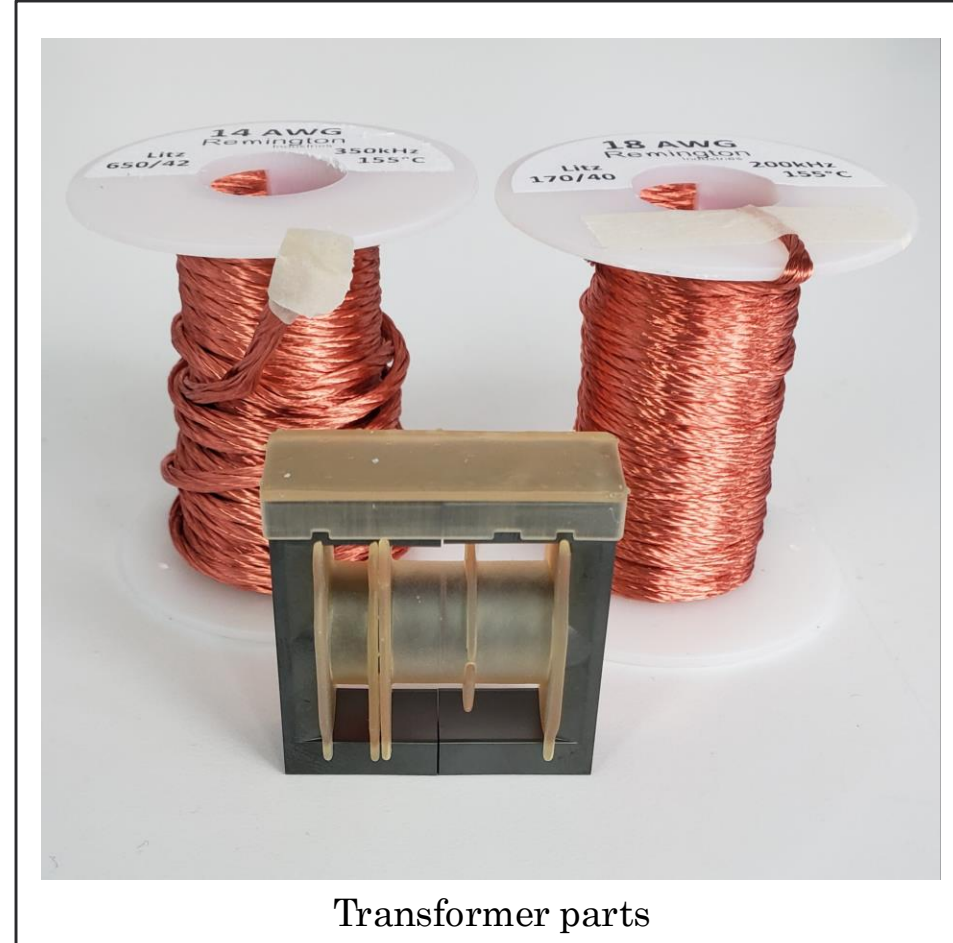
Control Loop

- Regulates the output voltage at 12 volts despite changes in input voltage and load
- If output voltage exceeds 12 V the switching frequency increases to lower the output voltage vice versa
- Feedback Loop:
 - Provides autonomous output regulation
 - Powered by output voltage so no external power needed



Transformer

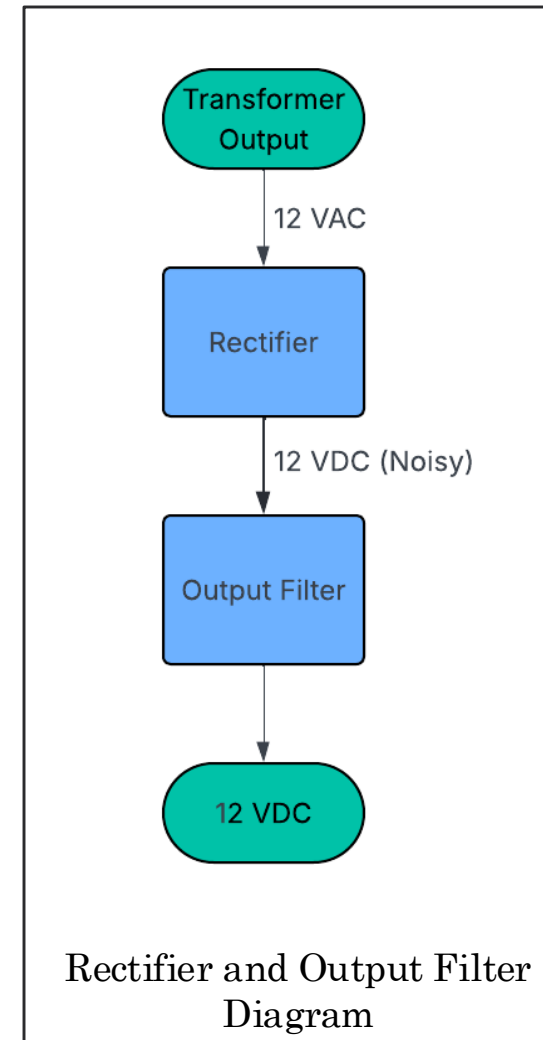
- The Transformer steps down AC voltage by a fixed ratio
- Our design has a transformer with 28:1 windings with a gap of 1mm
 - Reference design has 28:2 with a 1mm gap



Transformer parts

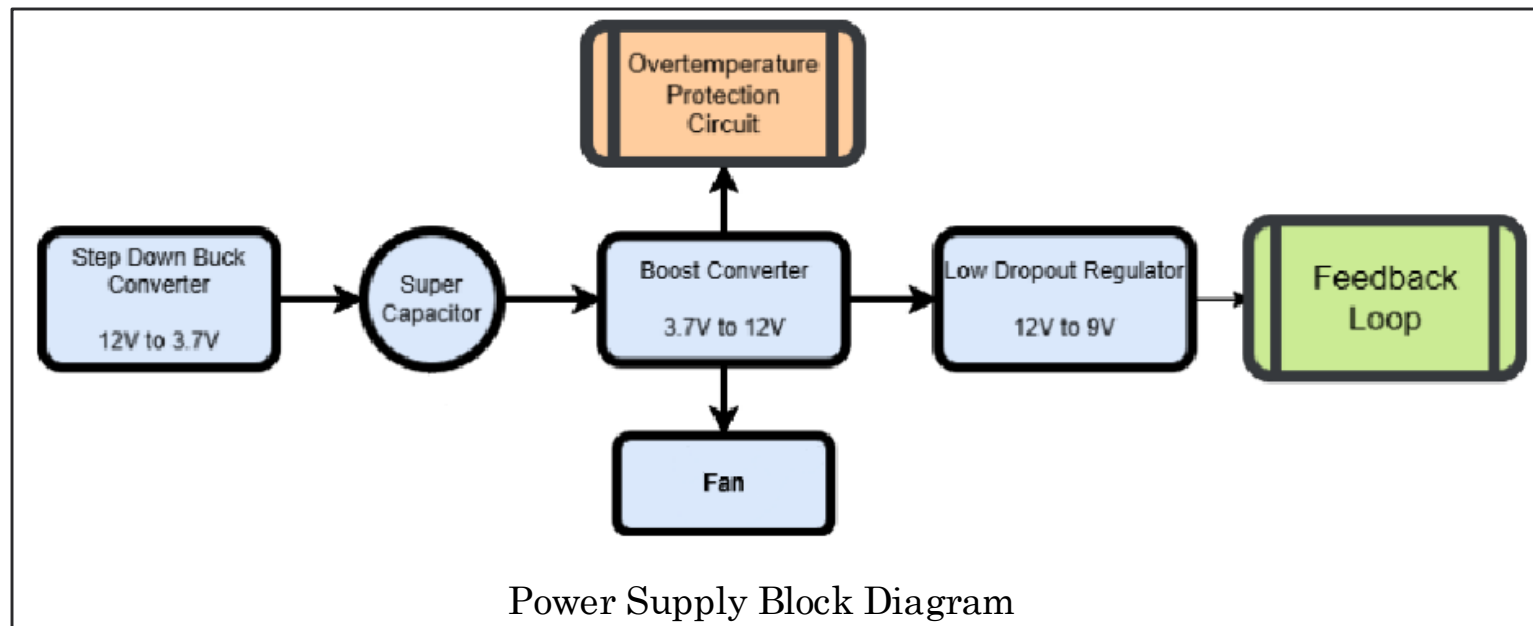
Rectifier and Output Filter

- Synchronous Rectifier:
 - Converts 12 VAC into 12 VDC
 - Lower conduction losses than diode rectifier
 - With a heat sink, MOSFETs' max current rating is 84 amps
- Output filter:
 - Smooths the output waveform, removing ripple
 - Low-pass LC filter



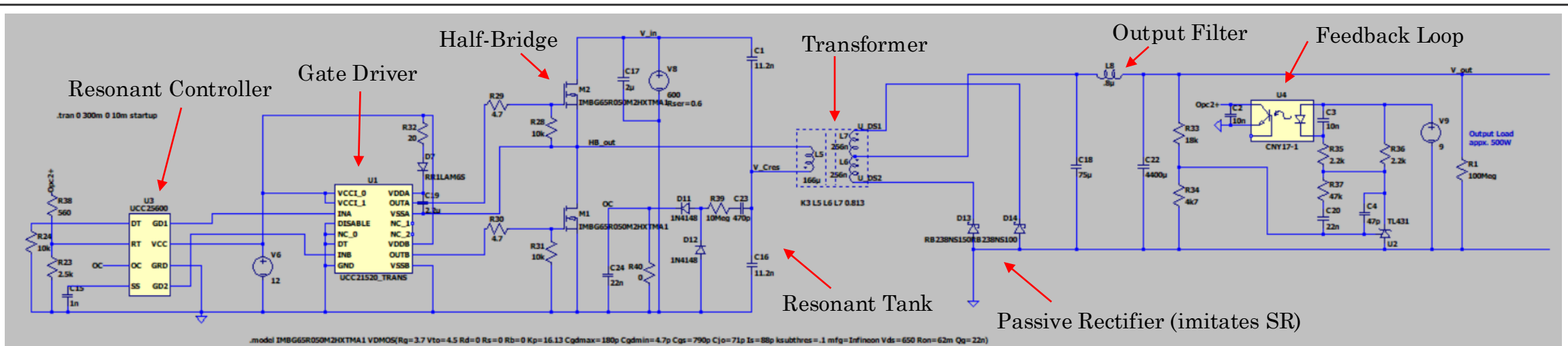
Power Supply

- Takes in the rectified voltage and uses a buck converter to step it down from 12 volts to 3.7 volts to charge a supercapacitor
- The supercapacitor acts as a small battery for the system
- Afterwards, a boost converter is used to step the voltage from 3.7 volts to 12 volts to supply power to the overtemperature protection circuit and port 1 of the fan voltage select
- Finally, there is a low dropout voltage regulator that drops the voltage from 12 volts to 9 volts to power the feedback loop



LTSPICE Simulation Summary

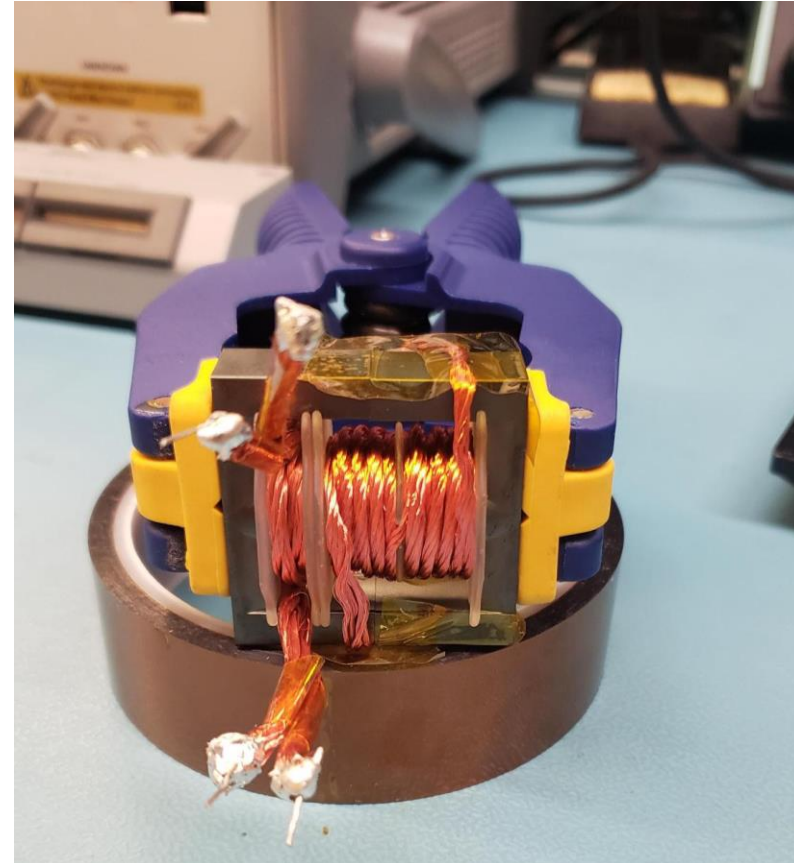
- Successfully simulated several essential elements of the system, including the half bridge, resonant tank, transformer, control loop, and output filter
- Simulated output line regulation, load regulation, and transient response
- Simulated overall efficiency of 97.7% (at about 368 watts of input power)



LTSPICE Main Simulation Circuit

Transformer Assembly

- Consists of four parts
 - Core
 - Bobbin
 - Primary & Secondary Windings
- Assembly
 - Transformer tuning was completed before, during, & after assembly
 - High temp resistant glue used to hold it together



Assembled Transformer

Board Assembly

- Occurred simultaneously with Transformer Assembly
- Majority of small SMD components were soldered by PCB vendor
- Large SMD and through-hole components were soldered by hand
- Cross-listed soldered components with BOM



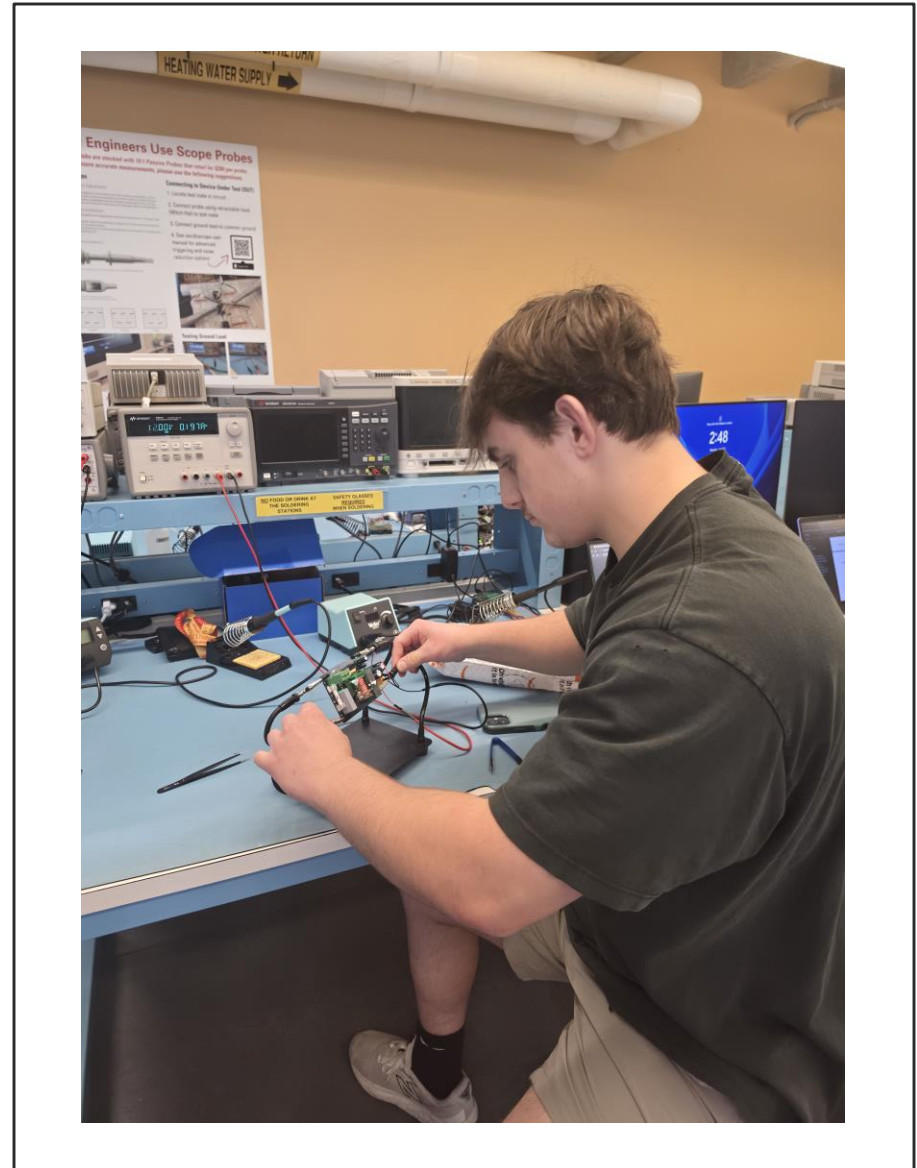
Top part of the board



Bottom part of the board

Testing Overview

- Unit Testing
 - Transformer
 - Isolation
 - Inductance
 - Board
 - Voltages
 - Waveforms
 - Currents
- System Testing
 - Phase 1: Primary Side Testing
 - Phase 2: Full System Testing
 - Tuning Efficiency and Input Range



Transformer Testing

- Inductance tuning
 - Measuring inductance of primary and secondary windings
 - Directly impacts efficiency of transformer & larger board design
- Isolation testing
 - Measuring if any voltage & current escapes the intended device/circuit
 - Transformer passed isolation test up to three times its rated voltage



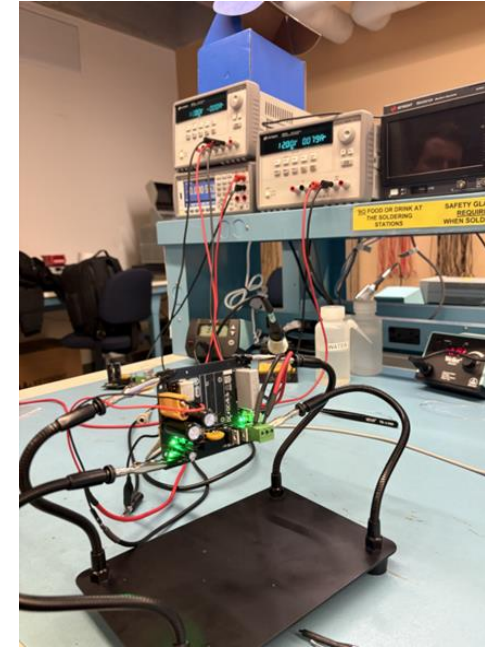
LCR Meter Inductance Reading



Isolation Tester 1.8 kV Pass

Board Testing

- Verified voltages at key areas
 - Startup Supercapacitor
 - Transformer
 - Control Loop
- Verified waveforms
 - Square waves at oscillator, gate driver, and MOSFET gates
- Observed current draw at low voltage side
 - 80mA draw at low voltage output
 - 200mA measured through power supply



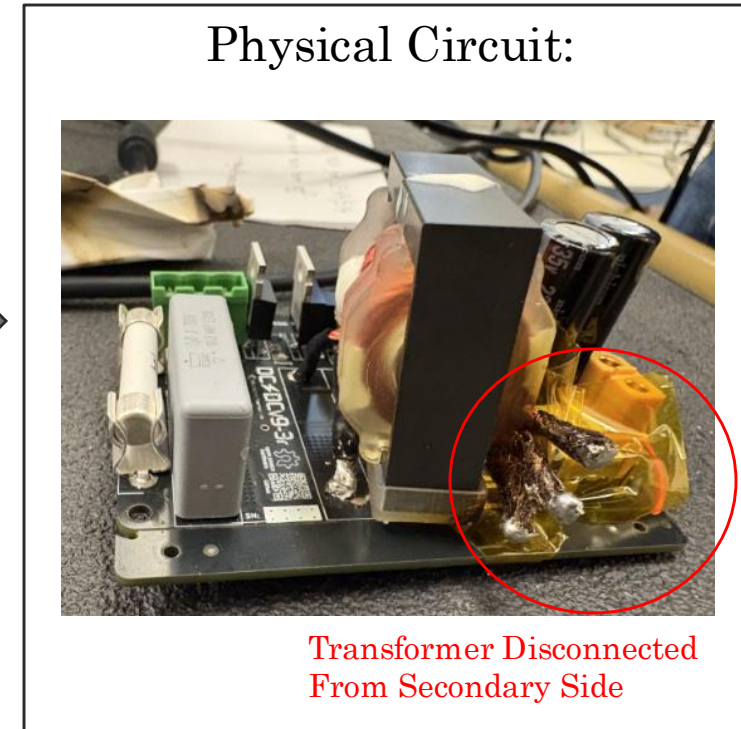
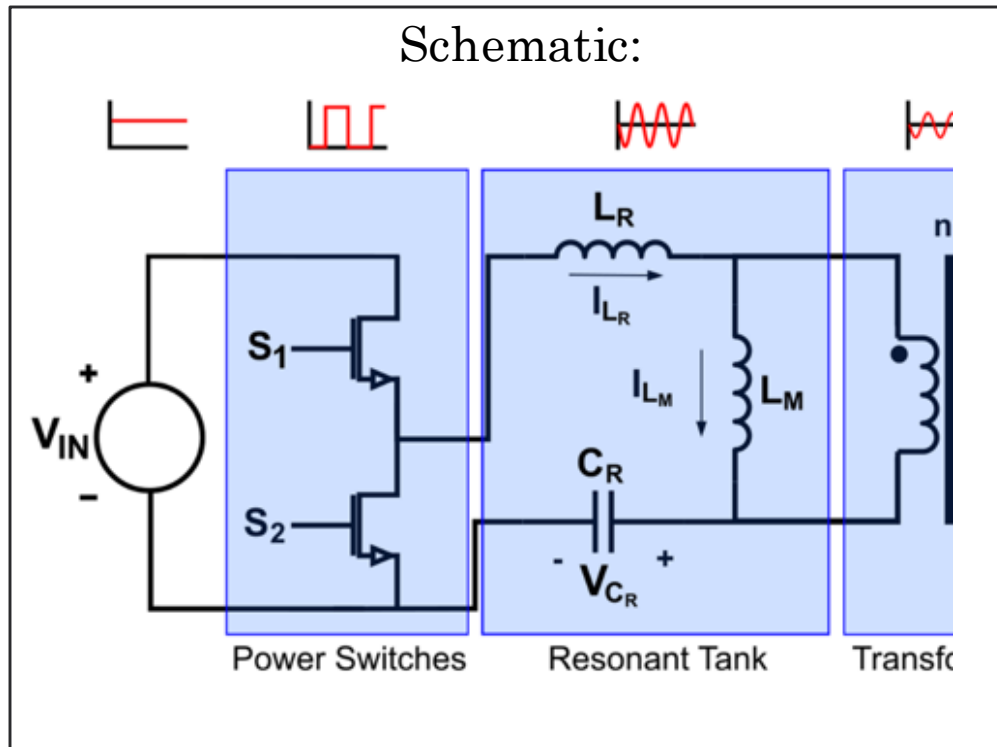
Board Testing Setup



Square Wave at Switching Network

System Testing

- Phase 1: Tested only primary side, ramping V_{IN} slowly from 0-580 volts



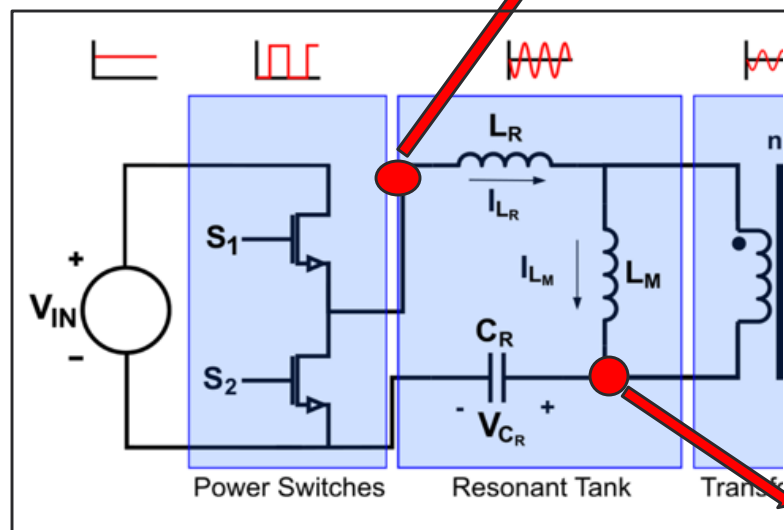
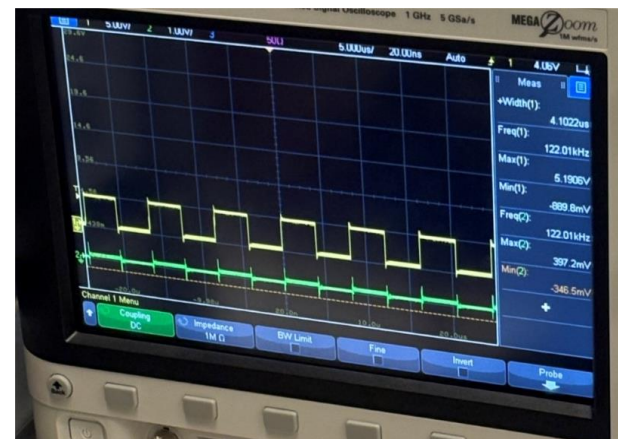
System Testing

Phase 1 Test Setup:

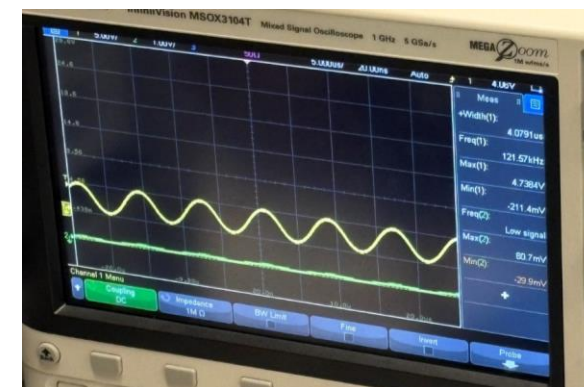


High Voltage Differential Probes
Used For Measurements

Square Wave Output From Half-Bridge:



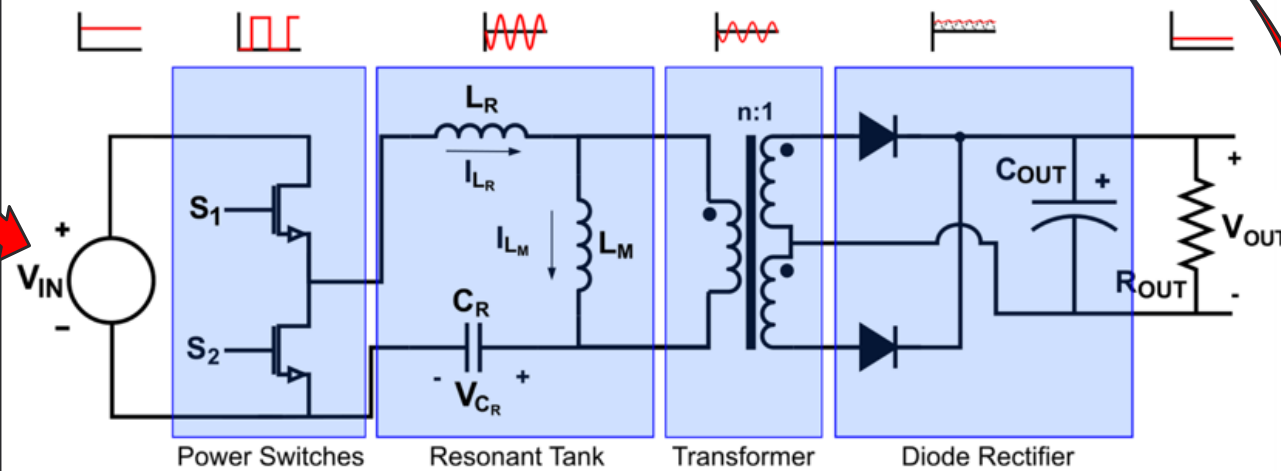
After Filtered By Resonant Tank:



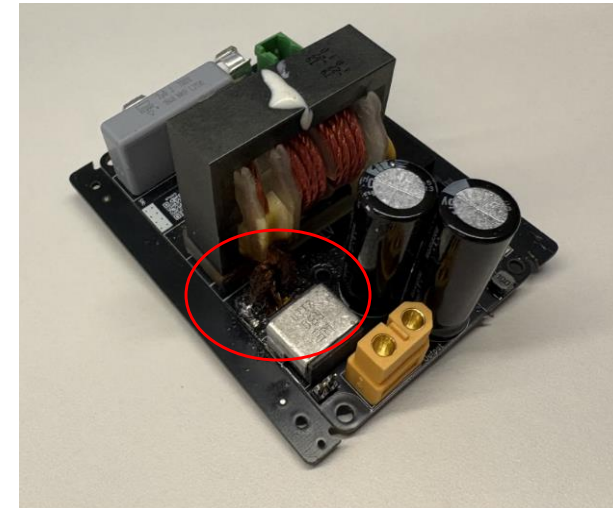
System Testing

- Phase 2: Connected transformer to secondary side to test entire system
 - Tested Output Line Regulation (vary input voltage)
 - Tested Output Load Regulation (vary load)

Schematic:



Physical Circuit:

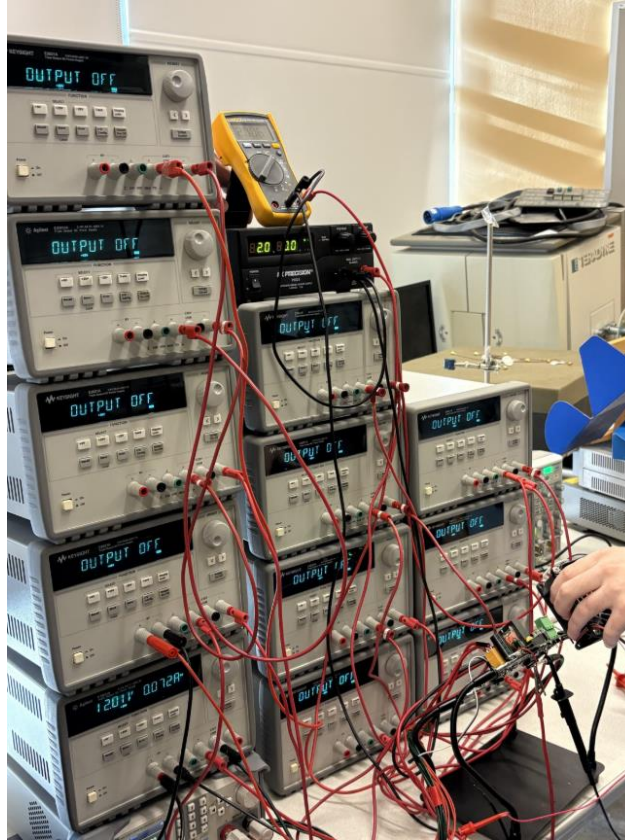


Transformer Connected To Secondary Side

System Testing

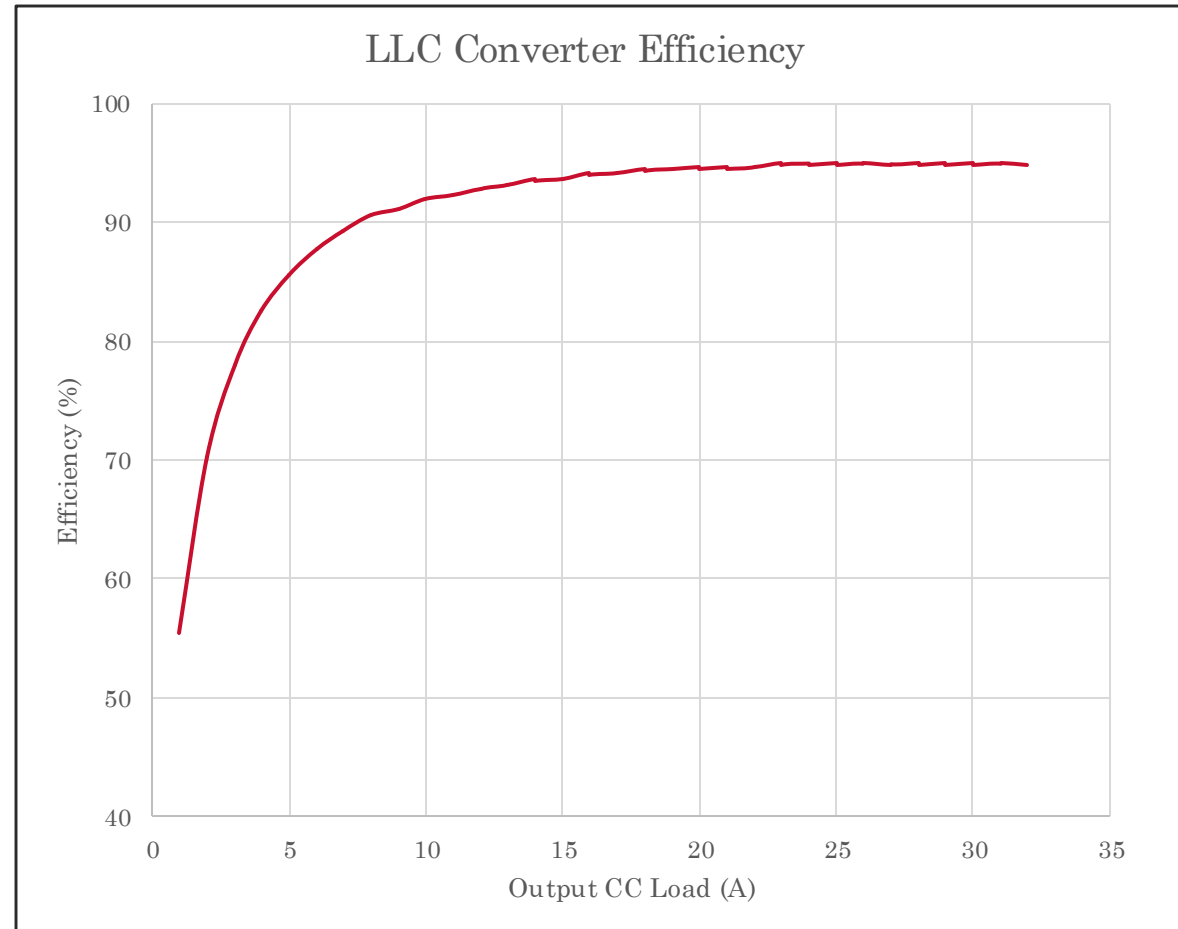
- Materials:
 - Input voltage: Twelve 50-volt power supplies in series
 - Load: Two DC-Electronic Loads in parallel
 - Multimeters used to observe inputs and output voltage
 - High Voltage Diff Probes
 - External fan used for temporary cooling
- Procedure:
 1. Ramp of V_{in} to 580 volts
 2. Increase output current from 0-32 amps

Phase 2 Test Setup:



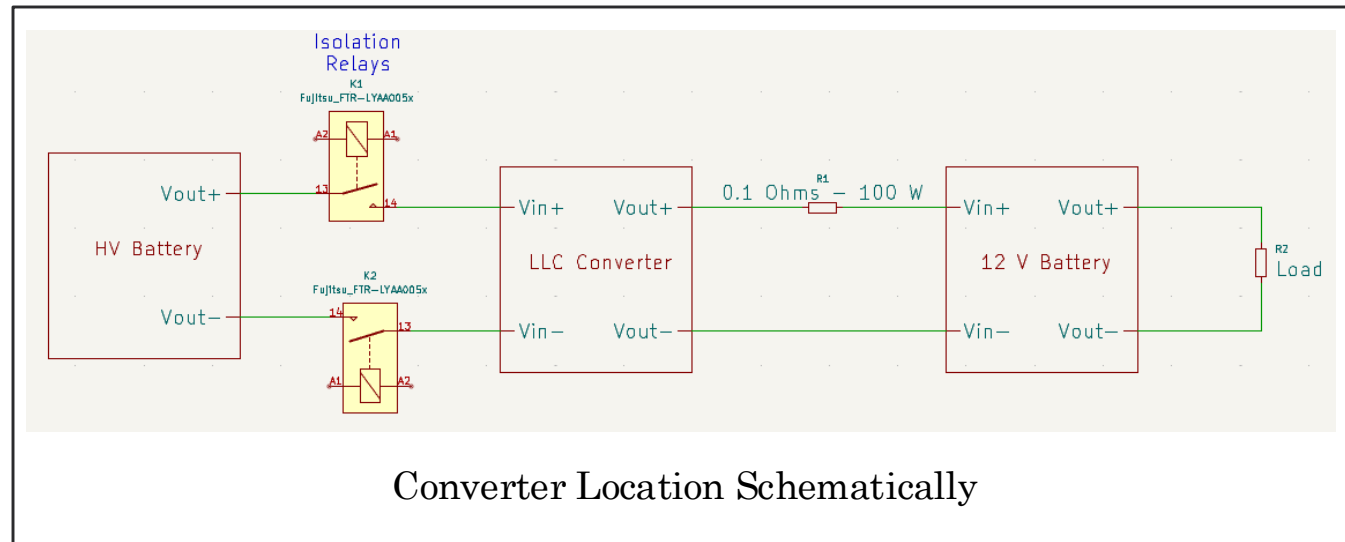
Results

- ✓ Peak Efficiency of 95%
- ✓ Stable 12 V output regulation for 350-600 V input voltage
- ✓ 32 A constant current load
- ✓ Galvanic Isolation
- ✓ Complies with FSAE regulations



Implementation

- Client's system changed mid-semester, and we adapted
- The converter will be implemented in the FSAE EV car to power their low voltage system as well as charge a low-voltage battery
- Modifications:
 - Supercapacitor no longer needed due to low-voltage battery (Much simpler power supply!)
 - No DC-enable logic necessary
 - External droop control resistor added for proper charging



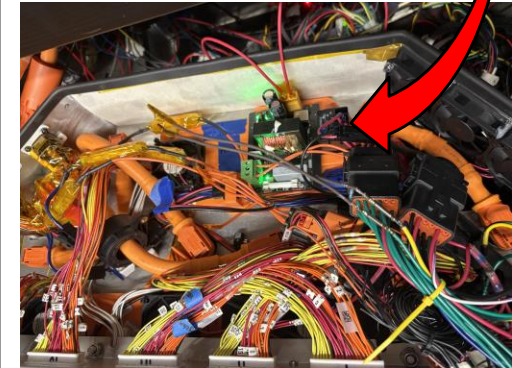
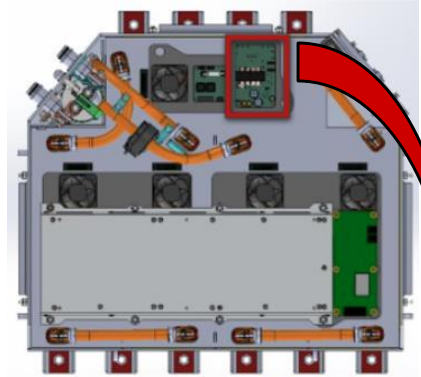
Implementation

- Unfortunately, the FSAE team was not ready to test with High-Voltage Battery until April 30th
- Current Issue:
 - High-Voltage Battery voltage sags from 580 volts to about 312 volts when loaded
 - Hence converter regulates the output to about only 11.23 volts, since its slightly below input voltage range
 - Most likely an issue on clients end

Formula Car:

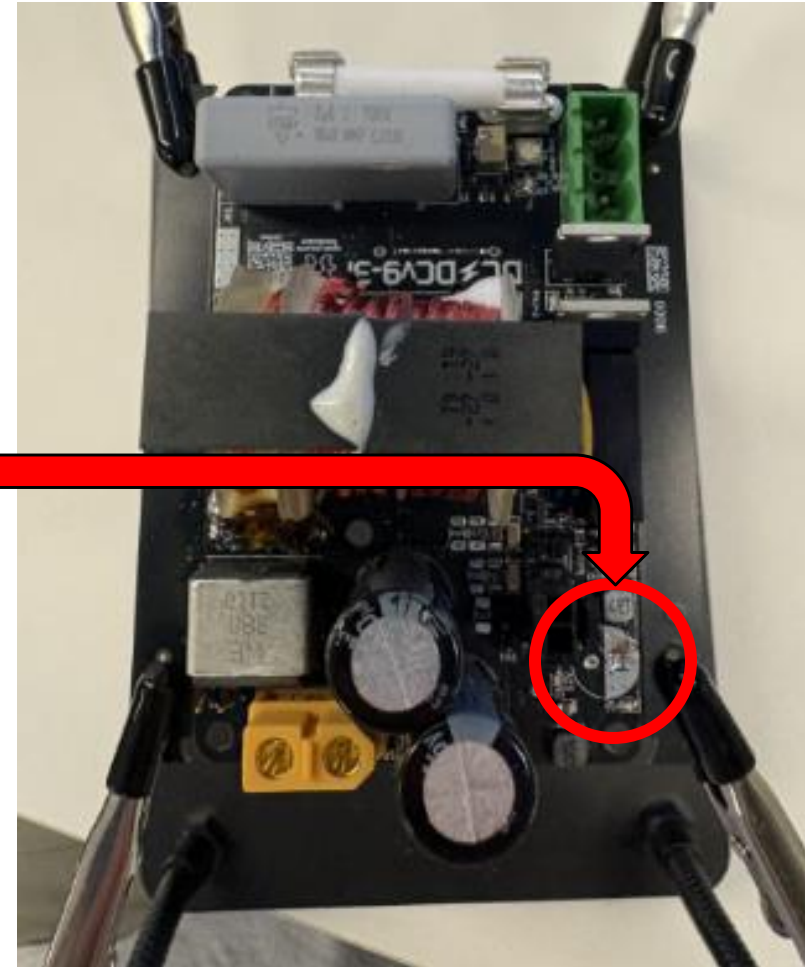


High Voltage Battery:



Update Dilemma

- The client made some last-minute changes by adding a 12-Volt battery to their car
- Recommended they didn't need it - they wanted it anyway
- Able to implement our converter with some slight changes
- In a larger scale context:
 - An additional battery would cause redundancy
 - Additional battery would cause more cost & pollution



Conclusion

- Converter exceeded all design requirements when tested in lab
- Implementation is still in progress due to pending issues with HV battery
- Learning Outcomes:
 - Power Electronics Experience (trial by fire!)
 - Troubleshooting (lots of it!)
 - Working on a complex system within a complex system
 - Transformer design and tuning (painful!)
 - Fundraising experience (Milwaukee Tool)
- Big Picture:
 - By using the converter, the FSAE team will be able to efficiently power car for several hours, opposed to swapping low voltage batteries every 20 minutes



Project Demo



Special Thanks!

Dr. Cheng Huang

Dr. Muhammad Rizwan Khan

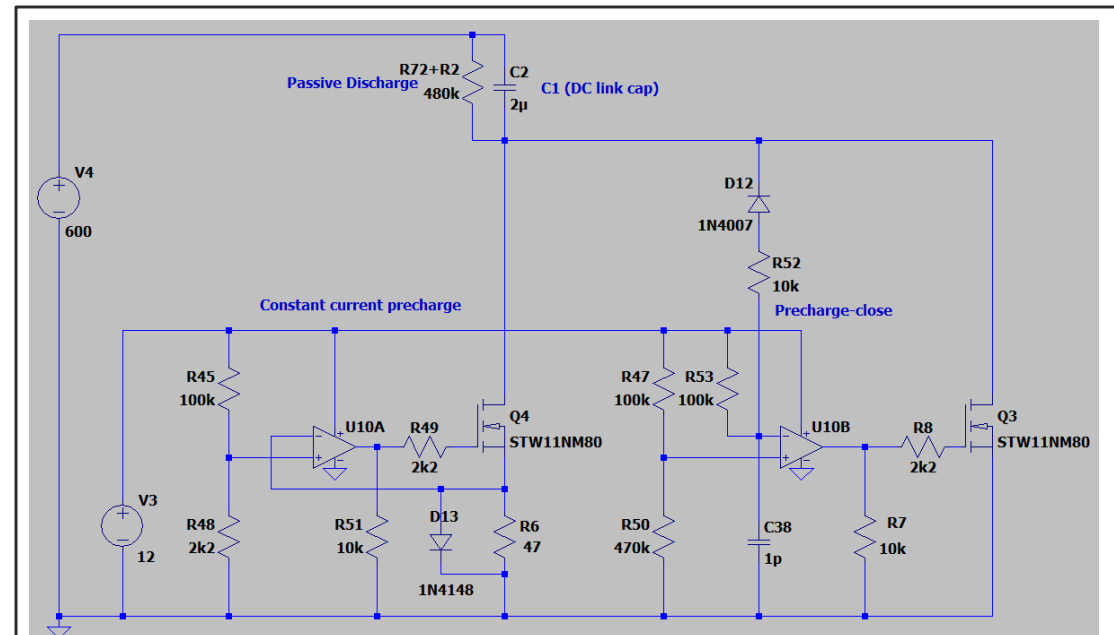
Thank you!

Appendix A

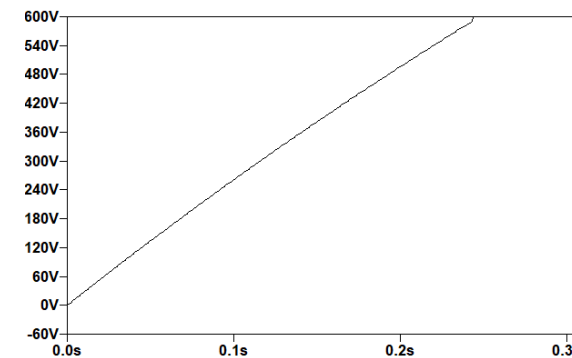
Simulation Results

Pre-Charge

- Used to pre-charge the DC-link capacitor to prevent a large inrush of current
- DC-link capacitor used to smooth out voltage from High Voltage Battery



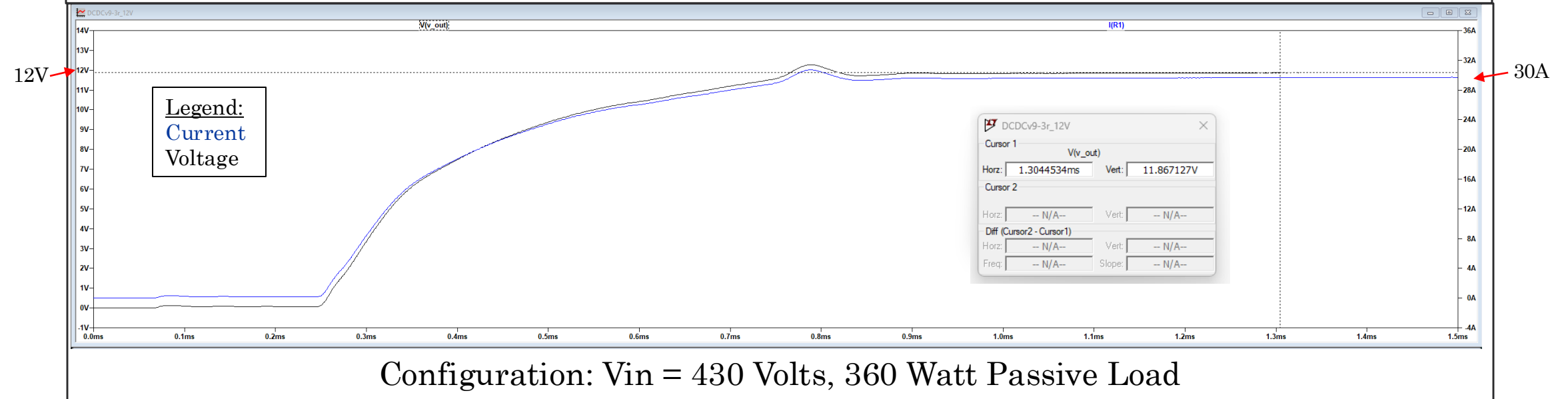
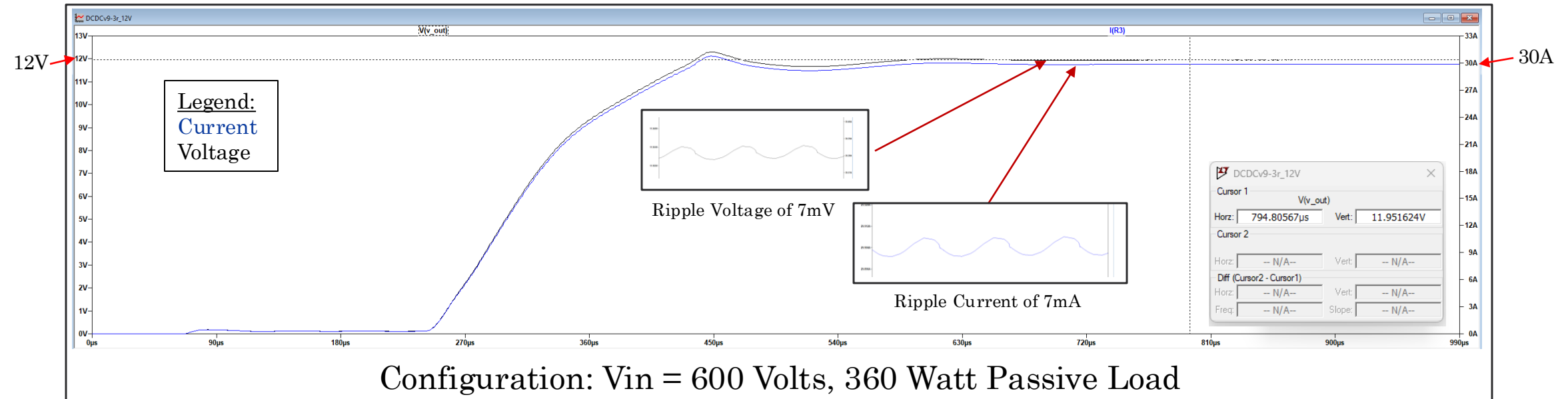
Precharge Circuit



Precharge Simulation

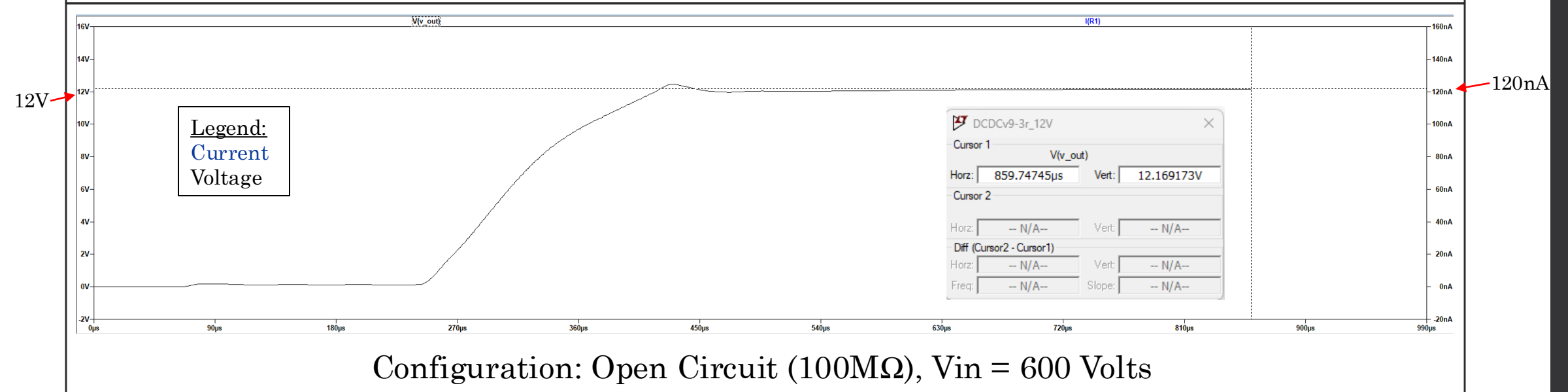
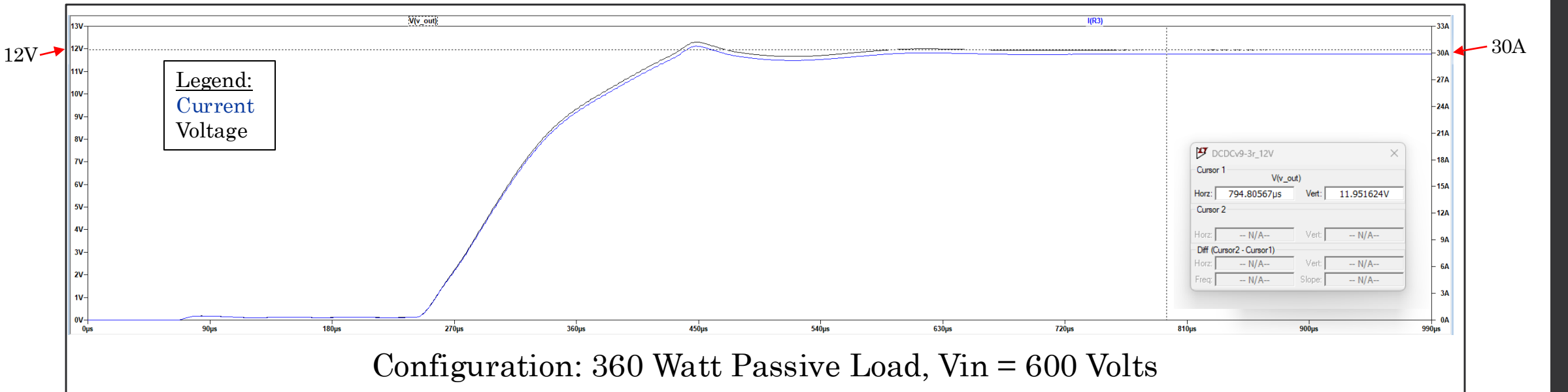
Output Line Regulation

Constant Load, Variable V_{in}

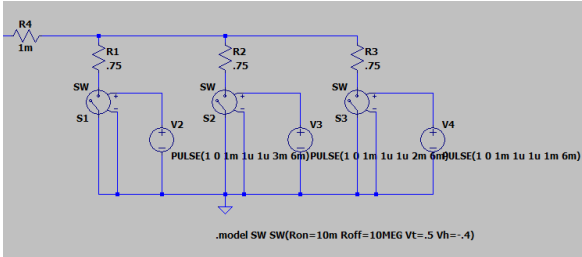


Output Load Regulation

Constant V_{in} , Variable Load

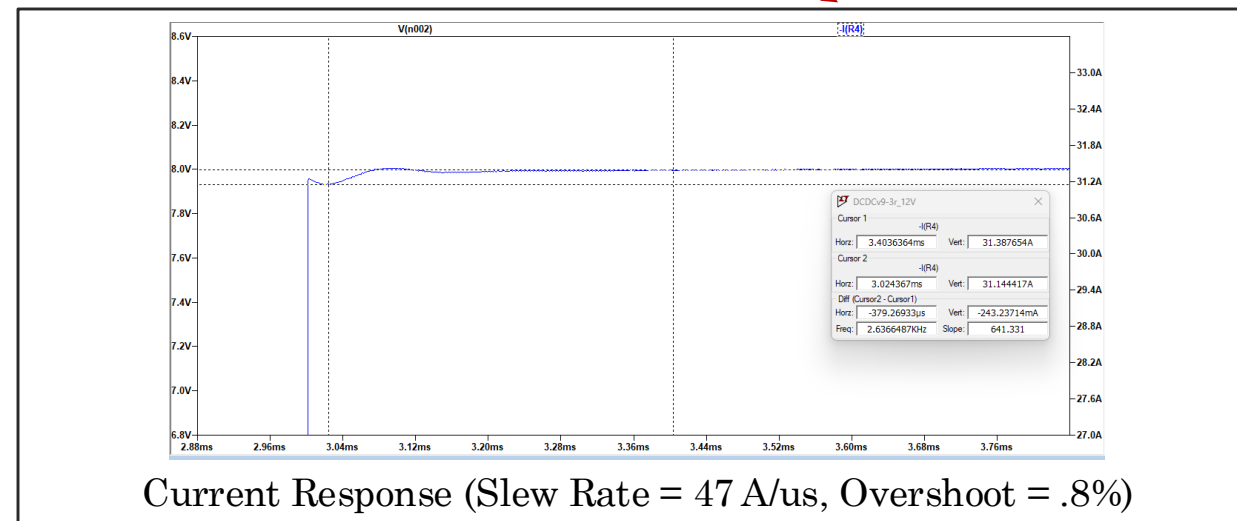
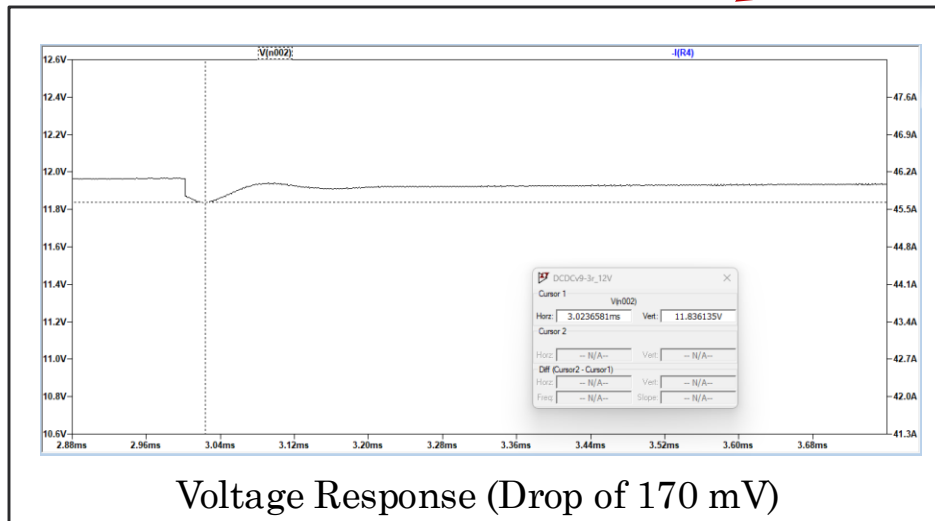
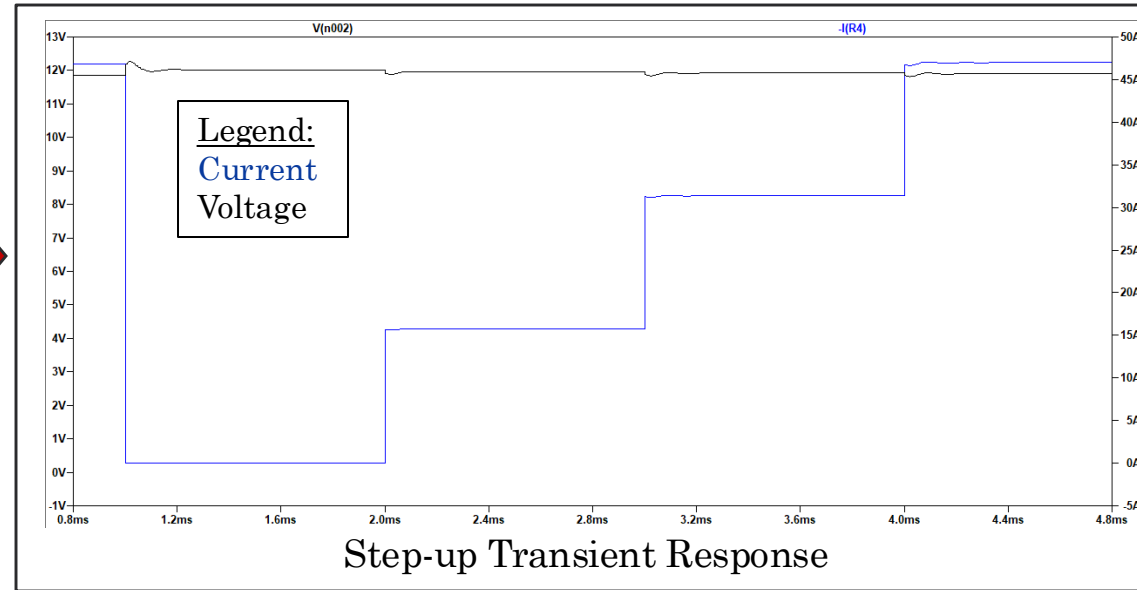


Transient Response



Dynamic Load SPICE Model

Load Step: 0→192→384→576 Watts
Time Step: 1 ms



Appendix B

Subsystems

Over-Temperature Protection

- Shuts down the fan when temperature above safe operating limit
- Vin+ connect to diode connected to op-amp output for latching
- Vin- connect to thermistor that uses a diode to measure max temp based on R36 and R35 resistors
- Fan control regulates fan speed by adjusting current or voltage flowing into it.

